



Today's agenda

- ↳ Single element 3
- ↳ Divide integers
- ↳ max AND pair



AlgoPrep



Single Number 3 $\rightarrow \{facebook, google, uber\}$

Given $arr[n]$, every element repeats twice except 2 elements. find the 2 unique elements.
 each occ. 1 time

Ex: $arr[i] = \{ \overset{0}{3} \overset{1}{6} \overset{2}{4} \overset{3}{4} \overset{4}{3} \overset{5}{8} \} \rightarrow \{6, 8\}$

$arr[4] = \{ \overset{0}{4} \overset{1}{9} \overset{2}{9} \overset{3}{8} \} \rightarrow \{4, 8\}$

Idea 1

use Hashmap and do counting of all the elements.

T.C: $O(n)$ S.C: $O(n)$

Idea 2

$arr[i] = \{ \overset{0}{3} \overset{1}{6} \overset{2}{4} \overset{3}{4} \overset{4}{3} \overset{5}{8} \}$

$$3^6^4^4^3^8 \rightarrow \cancel{3}^{\cancel{6}}^{\cancel{4}}^{\cancel{4}}^{\cancel{3}}^8 = 6^8 = 14$$

$\Downarrow$

14

14: $001110 \rightarrow$

0	1	1	0	0	$\rightarrow 18$
0	0	0	1	0	$\rightarrow 2$
					$\rightarrow 14$

arr[12] = 10 8 8 9 12 9 6 11 10 6 12 21

¹⁰¹⁰ ¹⁰⁰⁰ ¹¹⁰⁰ ⁰¹¹⁰ ¹⁰¹⁰ ¹¹⁰⁰
₁₀₀₀ ₁₀₀₁ ₁₀₀₁ ₀₁₀₁ ₀₁₁₀ ₁₀₁₀₁



ans: 11^21 = 30

$x:$
 $y:$
 $30:$ 1 1 1 1 0
 4 3 2 1 0

set 1 (1st bit index as 0)
 { 8 8 9 12 9 }
 12 21
 XOR 1 = 21

 set 2 (1st bit index 1)
 { 10 6 11 10 }
 6
 XOR 2 = 11

$0^1 = 1$ $0 \wedge 0 = 0$
 $1^0 = 1$ $1 \wedge 1 = 0$





//Pseudo Code

```
int singleNumber3 (int arr[N]) {
```

//Step 1

```
int v = 0;
```

```
for (int i = 0; i < N; i++) {
```

```
    v = v ^ arr[i];
```

```
}
```

v = 20

T.C: $O(N)$

S.C: $O(1)$

//Step 2

```
int idn = 0;
```

```
for (int i = 0; i < 32; i++) {
```

```
    if (checkBit(v, i) == true) {
```

```
        idn = i; break;
```

```
    }
```

//Step 3

```
int Set1 = 0; int Set2 = 0;
```

```
for (int i = 0; i < N; i++) {
```

```
    int num = arr[i];
```

```
    if (checkBit(num, idn) == true) {
```

```
        Set2 = Set2 ^ num;
```

```
    }
```

```
    else { Set1 = Set1 ^ num; }
```

```
}
```

```
Print (Set1);
```

```
Print (Set2);
```

}



//Pseudo code

//Step 1

```
int v=0;
for (int i=0; i<N; i++) {
    v = v ^ arr[i];
}
```

arr[12] = 10 8 8 9 12 9 6 11 10 6 12 21

1010 1000 1100 0110 1010 1100 1010 1010 1010 1010 1010 1010

1000 1001 1001 1001 1001 1001 1001 1001 1001 1001 1001 1001

$$v = 0 \oplus 10 \oplus 8 \oplus 8 \oplus 9 \oplus 12 \oplus 9 \oplus 6 \oplus 11 \oplus 10 \oplus 6 \oplus 12 \oplus 21$$

$$= 11 \oplus 21 = 30$$

$$v = 30 \rightarrow 11110$$

idx=0;

//Step 2

```
int idx=0;
for (int i=0; i<32; i++) {
    if (checkBit(v, i) == true) {
        idx = i; break;
    }
}
```

i=0 → checkBit(30, 0) → false

i=1 → checkBit(30, 1) → true

idx = 1

$$Set1 = 0 \oplus 8 \oplus 8 \oplus 9 \oplus 12 \oplus 9 \oplus 12 \oplus 21 = 21$$

$$Set2 = 0 \oplus 10 \oplus 6 \oplus 11 \oplus 10 \oplus 6 = 11$$

//Step 3

```
int Set1=0; int Set2=0;
for (int i=0; i<N; i++) {
    int num = arr[i];
    if (checkBit(num, idx) == true) {
        Set2 = Set2 ^ num;
    }
    else { Set1 = Set1 ^ num; }
}
```

arr[12] = 10 8 8 9 12 9 6 11 10 6 12 21

1010 1000 1100 0110 1010 1100 1010 1010 1010 1010 1010 1010

1000 1001 1001 1001 1001 1001 1001 1001 1001 1001 1001 1001

Q36 Max And Pair $\rightarrow$ Amazon



Q) Given $arr[N]$, find max $arr[i] \& arr[j]$ which can be obtained!
 $i \neq j$

Ex1: $arr[3] = \{27, 18, 20\}$

$27: 11011$

$27 \& 18 = 18 \rightarrow \text{ans}$

$18: 10010$

$18 \& 20 = 16$

$20: 10100$

$27 \& 20 = 16$

Ex2: $arr[5] = \{21, 18, 24, 17, 16\}$

$21: 10101$

$21 \& 16 = 16$

$16: 10000$

$21 \& 17 = 17 \rightarrow \text{ans}$

$17: 10001$

//idea 1

↳ check all Pairs and get the max out of it.

T.C: $O(N^2)$

S.C: $O(1)$

//idea 2

↳

$90000 > 89999$

$09000 < 89999$

$10010 > 01111$

↳ if leftmost is greater, whole number will be greater.



→ arr[i] = {26 13 23 28 7 25}

26: ^{4 3 2 1 0}
1 1 0 1 0

13: 0 1 1 0 1

23: 1 0 1 1 0

28: 1 1 1 0 0

7: 0 0 1 1 1

25: 1 1 0 0 1

Count: 4 3 1 1 1

010 = 0

011 = 0

110 = 0

111 = 1

26 & 28 = 24

28 & 25 = 24

26 & 25 = 24



//Pseudo Code

```
int maxAndPair (int arr[N]) {
```

```
    for (int i=31; i>=0; i--) {
```

```
        int c=0;
```

```
        for (int j=0; j<N; j++) {
```

```
            if (checkBit(arr[j], i) == true) {  
                c++;  
            }
```

```
            if (c >= 2) {
```

```
                for (int j=0; j<N; j++) {
```

```
                    if (checkBit(arr[j], i) == false) {  
                        arr[j] = 0;  
                    }
```

T.C: $O(32 \times 2N)$
 $\approx O(62N) \approx O(N)$
S.C: $O(1)$

```
        int idn1 = -1;
```

```
        int idn2 = -1;
```

```
        for (int k=0; k<N; k++) {
```

```
            if (arr[k] > 0 && idn1 == -1) {  
                idn1 = k;
```

```
            }  
            else if (arr[k] > 0 && idn1 != -1) {  
                idn2 = k;  
                break;  
            }
```

return arr[idn1] & arr[idn2];



2

→ arr[i] = {26 13 23 28 7 25}

for (int i=31; i>=0; i--)

int c=0;

for (int j=0; j<N; j++)

if (checkBit(arr[j], i) == true)

c++;

3

if (c>=2)

for (int j=0; j<N; j++)

if (checkBit(arr[j], i) == false)

arr[j] = 0;

3

int idn1 = -1;

int idn2 = -1;

for (int k=0; k<N; k++)

if (arr[k] > 0 & idn1 == -1)

idn1 = k;

else if (arr[k] > 0 & idn1 != -1)

idn2 = k;

break;

3

idn1 = 1 0

idn2 = 1 3

26 0 1 1 0 1 0

0 1 1 3 0 0 1 0 0

2 2 3 0 0 0 1 0

2 3 2 8 1 1 1 0 0

4 7 0 0 1 0 1 0

5 2 5 1 1 0 0 1

c = 0 1 2 3

26 & 28



Q) Divide integers

↳ Divide two integers without using multiplication, division or mod operators.

$$\text{Ex: } 30/7 = 4$$

$$48/5 = 9$$

$$5/2 = 2$$

Idea 1

$$48 \quad 5$$
$$+ 48 - 5 = 43$$

$$42 - 5 = 37$$

$$37 - 5 = 32$$

$$32 - 5 = 27$$

$$27 - 5 = 22$$

$$22 - 5 = 17$$

$$17 - 5 = 12$$

$$12 - 5 = 7$$

$$7 - 5 = 2$$

T.C: $O(\log)$

S.C: $O(1)$

ans = 9



1/idea2

$$a = 48$$

$$b = 5$$

$$ans = 9$$

$$9 = \begin{array}{cccccc} & 4 & 3 & 2 & 1 & 0 \\ - & - & 0 & 0 & 0 & 1 \\ & & 2^3 & & & 2^0 \end{array} = 9$$

$$temp = 0 + 40 + 5$$



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//Pseudo code

```
int divideIntegers (int A, int B){  
    long a = A;  
    long b = B;
```

```
    int Sign = 1;  
    if (A < 0) { Sign = Sign * -1; }  
    if (B < 0) { Sign = Sign * -1; }  
    a = Math.abs(a);  
    b = Math.abs(b);
```

```
    int q = 0;  
    int temp = 0;
```

```
    for (int i = 31; i >= 0; i--) {  
        if (temp + (b << i) <= a) {  
            temp = temp + (b << i);  
            q = q + ((long)1 << i);  
        }  
    }
```

```
    if (Sign < 0) { q = -q; }
```

```
    if (q > Integer.MAX_VALUE) {  
        return Integer.MAX_VALUE;  
    }  
    if (q < Integer.MIN_VALUE) {  
        return Integer.MIN_VALUE;  
    }
```

T.C: $O(32)$

S.C: $O(1)$

```
3 return (inp);
```



tracing

```
long a = A;
long b = B;
```

```
int Sign = 1;
if (A < 0) { Sign = Sign * -1; }
if (B < 0) { Sign = Sign * -1; }
a = Math.abs(a);
b = Math.abs(b);
```

```
int q = 0;
int temp = 0;
```

```
for (int i = 31; i >= 0; i--) {
    if (temp + (b << i) <= a) {
        temp = temp + (b << i);
        q = q + ((long)1 << i);
    }
}
```

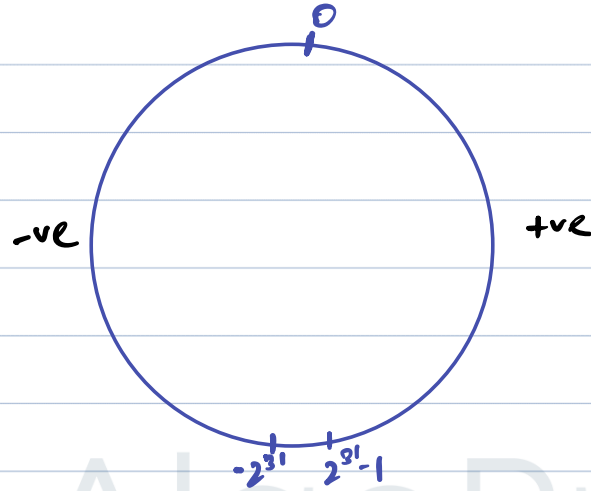
$a = 95$ $b = 5$ $\Rightarrow \text{ans} = 19$
 visualization: 2^31 5^4 3^3 2^2 1^1 0^0
 $q = 0 + 16 + 2 = 18 + 1 = 19$

$\text{temp} = 0 + 80 + 10 + 5$



* Numbers in DSA

$$\text{int} \rightarrow \{-2^{31}, 2^{31}-1\}$$



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